

4 A coral reef is an underwater ecosystem.

The photograph shows a coral reef in the Caribbean.



(Source: © Belize / Alamy Stock Photo)

A photosynthetic bacterium found on these coral reefs produces a chemical called curacin A.

This chemical has been found to be effective against some colon and kidney cancers.

- (a) Large quantities of this organism need to be grown in a culture medium in order to extract enough curacin A.

The culture is kept at the optimum temperature for the growth of this bacterium.

Suggest **two** other conditions that would be needed for maximum growth of this bacterium.

(2)

1

2

- (b) Scientists conduct experiments to determine the effectiveness and safety of new drugs.

- (i) Which of the following scientists tested the effectiveness of digitalis soup?

(1)

- ☐ **A** Adolf Fick
- ☐ **B** Matthew Meselson and Franklin Stahl
- ☐ **C** Godfrey Hardy and Wilhelm Weinberg
- ☐ **D** William Withering



- (ii) Curacin A is being developed as a drug to treat some colon and kidney cancers in humans.

Suggest how a suitable dose for cancer treatment would be determined in human trials.

(2)

- (iii) Describe how a double-blind clinical trial would be performed with this cancer drug.

(4)

(Total for Question 4 = 9 marks)



P 6 7 1 0 3 A 0 9 2 8